

MD 2 – CAD Task

1 Introduction

Before working on this assignment, we recommend that you work on the exercise sheets. The exercise sheets can be found in the Ilias course “CAD at IPEK”. We wish you good luck with your work. For questions or comments on the assignment, please contact us at cad@ipek.kit.edu.

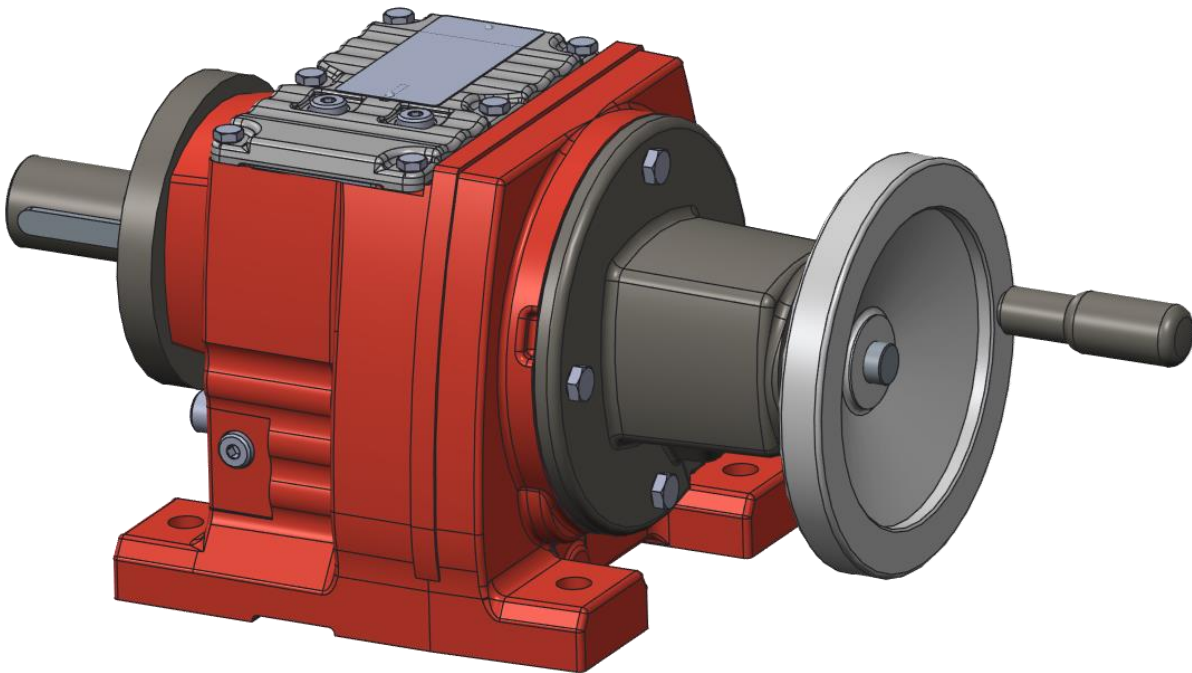


Figure 1: Complete model of the gearbox

For the certificate, your task is to edit a CAD model of the three-stage gearbox that you already know from the MD 1 workshops.

The CAD models created as part of the following sub-tasks serve as a basis for discussion and demonstrations and must therefore be processed **completely** before your certificate examination.

We recommend that you set the **working directory** to the folder with your files each time you start Creo.

2 Summary of the sub-tasks

1. Open the model "GEAR_ASSEMBLY.ASM" in Creo 8.0.
ASSEMBLY - Environment
2. Create the seal holder for the flange of the output shaft.
PART - Environment
3. Add the seal holder and the missing parts to the gearbox assembly.
ASSEMBLY - Environment
4. Create the missing output shaft as a new part.
PART - Environment
5. Create a sub-assembly for the output shaft and place it in the main assembly.
ASSEMBLY - Environment
6. **Create section views of the completed assembly, which contains cuts of all four shafts**
TOOLS - View manager
7. **Create a Simplified Representation containing no screws**
TOOLS - View manager
8. **Create a drawing of the input shaft**
DRAWING - Environment

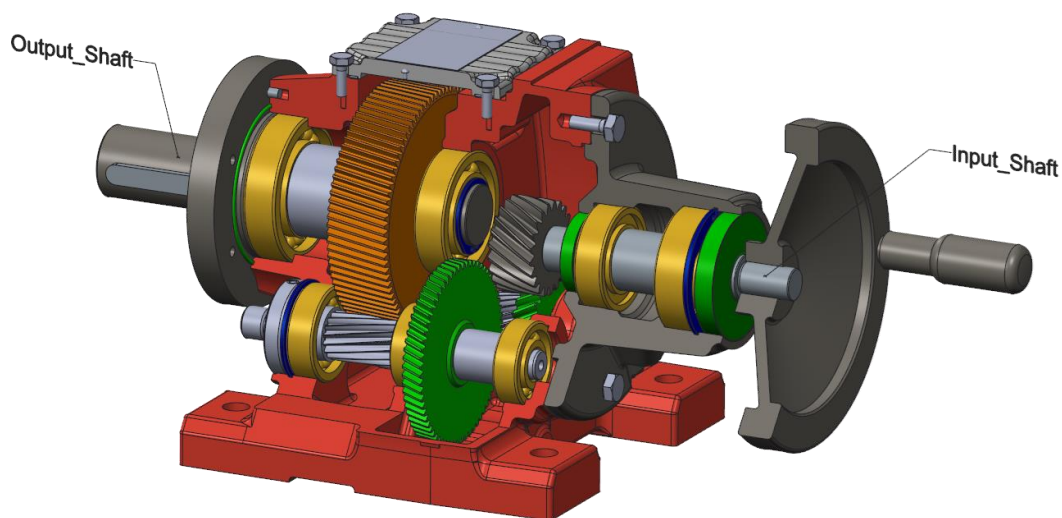


Figure 2: Section view of the gearbox

2.1 Task 1

Open the assembly model "GEAR_ASSEMBLY.ASM" in Creo 8.0.

The model parts are provided on Ilias as a zip packed folder named "assembly_parts.zip". This folder contains the main assembly model and all preexisting parts and sub-assemblies. In order to avoid troubles saving the files, it is best to store them in a common folder. From this folder, open the assembly model "GEAR_ASSEMBLY.ASM" in Creo 8.0.

Note:

- Unzip all the parts into one folder
- Use section views to display internal parts in the assembly

2.2 Task 2

Create the seal holder for the flange of the output shaft.

The seal holder serves multiple purposes: Firstly, it hosts the radial shaft sealing as well as an O-ring and secondly, it fixes the outer ring of the output shaft's locating bearing.

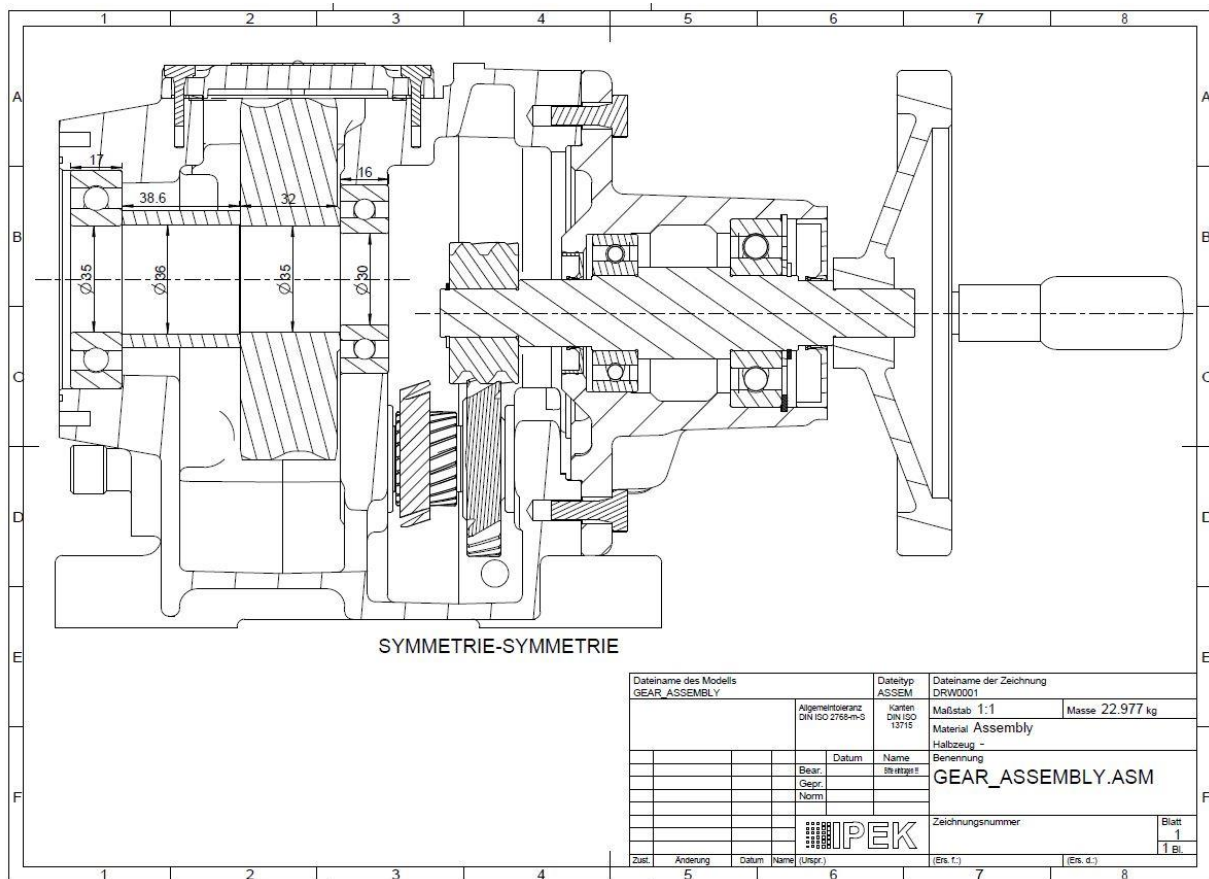


Figure 3: Section view of the gearbox as technical drawing

To create the seal holder, use the dimensions specified in the technical drawing “Installation drawing” (figure 5) at the end of this document. The remaining dimensions can be obtained directly from the CAD model via the measure tool (“Analysis” > “Measure”). Name the new part “SEAL HOLDER OUTPUT SHAFT.PRT”.

Note: Remember, the seal holder has to include a groove for the O-ring as well as a radial shaft seal. In addition to the lecture notes, use common literature (KIT library) for further information on the installation of seals.

2.3 Task 3

Add the parts “SEAL HOLDER OUTPUT SHAFT.PRT”, “RADIALWELLENDICHTRING_62.PRT” (radial shaft sealing), “O-RING.PRT” and “M5X20.PRT” to the assembly “FLANGE.ASM”.

2.4 Task 4

Create a new part called "OUTPUT_SHAFT.PRT". Use the dimensions specified in the technical drawing "Installation drawing" at the end of this document.

The remaining dimensions can be obtained directly from the CAD model via the measure tool.

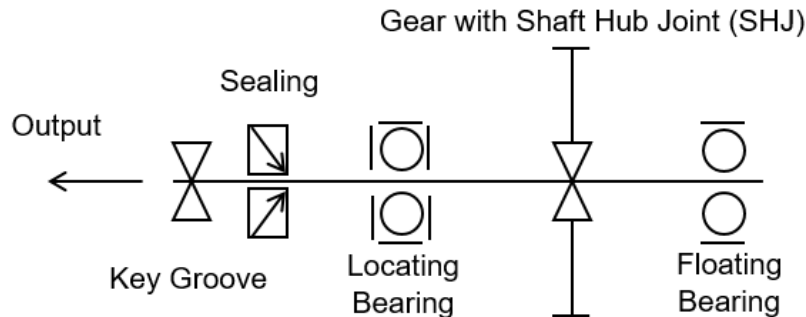


Figure 4: Principle sketch of the output shaft

2.5 Task 5

After modeling the output shaft, create a new sub-assembly named "OUTPUT_SHAFT_SUBASSY.ASM".

Gear – Model Tree (p. 5-6) shows the parts that "OUTPUT_SHAFT_SUBASSY.ASM" shall contain. Add this sub-assembly to the main assembly "GEAR_ASSEMBLY.ASM".

Moreover, add the parts "SICHERUNGSRING_29_36.PRT" (circlip) and "PASSFEDER_R5_8_46.PRT" (feather key) to the main assembly.

2.6 Task 6

Create Section Views in your main assembly (one or more), so that all four gear shafts are "sectioned". Exclude components like shafts and screws from your section(s).

2.7 Task 7

Define a Simplified Representation in the View Manager, which excludes the display of screws in the main assembly.

2.8 Task 8

Create a drawing of the input shaft "INPUT_SHAFT.PRT".

Pay attention to the representation of the shaft in different views (full model, sectional view), as well as the specification of dimensions and tolerances. The tolerance values may be chosen freely, references can be found in the literature.

Please note that for the successful submission of the drawing at least 3 views should be displayed (one of which should be a full section view), and all necessary dimensions and information like name, date and a telephone number (please fill in your matriculation number here) should also be included.

*Information on this can be found on exercise sheets seven and eight in the Ilias course "CAD at IPEK". Please bring the drawing printed out for the **certificate examination**. Pay attention to the "Actual size" setting so that your dimensions are displayed true to scale.*

3 Gear – Model Tree

 GEAR_ASSEMBLY.ASM	(in Main-Assembly)
 HOUSING.ASM	(in Main-Assembly, complete)
 6KT_SCHRAUBE_M5X16.PRT	(in Sub-Assembly)
 6KT_SCHRAUBE_1_M5X16.PRT	(in Sub-Assembly)
 6KT_SCHRAUBE_3_M5X16.PRT	(in Sub-Assembly)
 6KT_SCHRAUBE_M6X16.PRT	(in Sub-Assembly)
 6KT_SCHRAUBE_1_M6X16.PRT	(in Sub-Assembly)
 6KT_SCHRAUBE_2_M6X16.PRT	(in Sub-Assembly)
 6KT_SCHRAUBE_M8X20.PRT	(in Sub-Assembly)
 6KT_SCHRAUBE_1_M8X20.PRT	(in Sub-Assembly)
 6KT_SCHRAUBE_6-4X16.PRT	(in Sub-Assembly)
 6KT_SCHRAUBE_1_6-4X20.PRT	(in Sub-Assembly)
 DECKEL.PRT	(in Sub-Assembly)
 DECKEL_1.PRT	(in Sub-Assembly)
 DECKEL_2.PRT	(in Sub-Assembly)
 DECKEL_KLEIN.PRT	(in Sub-Assembly)
 DECKEL_K.PRT	(in Sub-Assembly)
 DECKEL_DICHTUNG.PRT	(in Sub-Assembly)
 DICHTKAPPE.PRT	(in Sub-Assembly)
 ENTLUEFTUNGSVENTIL.PRT	(in Sub-Assembly)
 RILLENKUGELLAGER.PRT	(in Sub-Assembly)
 RIKULA_30_62.PRT	(in Sub-Assembly)
 RIKULA_47.PRT	(in Sub-Assembly)
 RIKULA_15_35.PRT	(in Sub-Assembly)
 RIKULA_25_52.PRT	(in Sub-Assembly)
 RIKULA_62_25.PRT	(in Sub-Assembly)
 RIKULA_42.PRT	(in Sub-Assembly)
 RIKULA_35_15.PRT	(in Sub-Assembly)
 RIKULA_42_20.PRT	(in Sub-Assembly)
 RIKULA_35_72.PRT	(in Sub-Assembly)
 RIKULA_INNENRING.PRT	(in Sub-Assembly)
 PASSFEDER_5X5X10.PRT	(in Sub-Assembly)
 PASSFEDER_D5_5_20.PRT	(in Sub-Assembly)
 PASSFEDER_8_10_28.PRT	(in Sub-Assembly)
 PASSFEDER_6_6_10.PRT	(in Sub-Assembly)
 SICHERUNGSRING_54.PRT	(in Sub-Assembly)
 SICHERUNGSRING_30_2_37.PRT	(in Sub-Assembly)
 SICHERUNGSRING_30-2.PRT	(in Sub-Assembly)
 SICHERUNGSRING_15-2.PRT	(in Sub-Assembly)
 SICHERUNGSRING_24.PRT	(in Sub-Assembly)
 SICHERUNGSRING_41.PRT	(in Sub-Assembly)
 VERSCHLUSS.PRT	(in Sub-Assembly)
 VERSCHLUSS_1.PRT	(in Sub-Assembly)
 VERSCHLUSS_2.PRT	(in Sub-Assembly)
 VERSCHLUSS_3.PRT	(in Sub-Assembly)
 VERSCHLUSSKAPPE.PRT	(in Sub-Assembly)
 VERSCHLUSSKAPPE_1.PRT	(in Sub-Assembly)
 HANDRAD.PRT	(in Sub-Assembly)

	HANDRAD_SCHEIBE.PRT	(in Sub-Assembly)
	RITZELWELLE.PRT	(in Sub-Assembly)
	RITZELWELLE_3.PRT	(in Sub-Assembly)
	RADIALWELLENDICHTRING.PRT	(in Sub-Assembly)
	RADIALWELLENDICHTRING_21-7_62.PRT	(in Sub-Assembly)
	ZAHNRAD.PRT	(in Sub-Assembly)
	ZAHNRAD_120_35.PRT	(in Sub-Assembly)
	ZAHNRAD_16_47.PRT	(in Sub-Assembly)
	ZAHNRAD_16_65.PRT	(in Sub-Assembly)
	DISTANZROHR.PRT	(in Sub-Assembly)
	ANTRIEBSDECKEL.PRT	(in Sub-Assembly)
	ABSTANZHUELSE_36_45.PRT	(in Sub-Assembly)
	PLATTE.PRT	(in Sub-Assembly)
	LAGERVERSCHLUSS.PRT	(in Sub-Assembly)
	UNTERLEGSCHEIBE.PRT	(in Sub-Assembly)
	GETRIEBEGEHAEUSE.PRT	(in Sub-Assembly)
	INPUT_SHAFT.ASM	(in Sub-Assembly)
	INPUT_SHAFT.PRT	(in Sub-Assembly)
	PASSFEDER_R2_4_14.PRT	(in Sub-Assembly)
	FLANGE.ASM	(not Sub-Assembly)
	SEAL HOLDER_OUTPUT_SHAFT.PRT	(not Sub-Assembly)
	O_RING.PRT	(in Folder)
	RADIALWELLENDICHTRING_62.PRT	(in Folder)
	Muster 1 von M5x20.PRT	(not Sub-Assembly)
	M5x20.PRT	(in Folder)
	M5x20.PRT	(in Folder)
	M5x20.PRT	(in Folder)
	M5x20.PRT	(in Folder)
	M5x20.PRT	(in Folder)
	M5x20.PRT	(in Folder)
	OUTPUT_SHAFT_SUBASSY.ASM	(not in Sub-Assembly)
	OUTPUT_SHAFT.PRT	(not in Sub-Assembly)
	SICHERUNGSRING_29_36.PRT	(in Folder)
	PASSFEDER_R5_8_46.PRT	(in Folder)

Note: Save all parts in one folder

Don't save any model in a separate folder

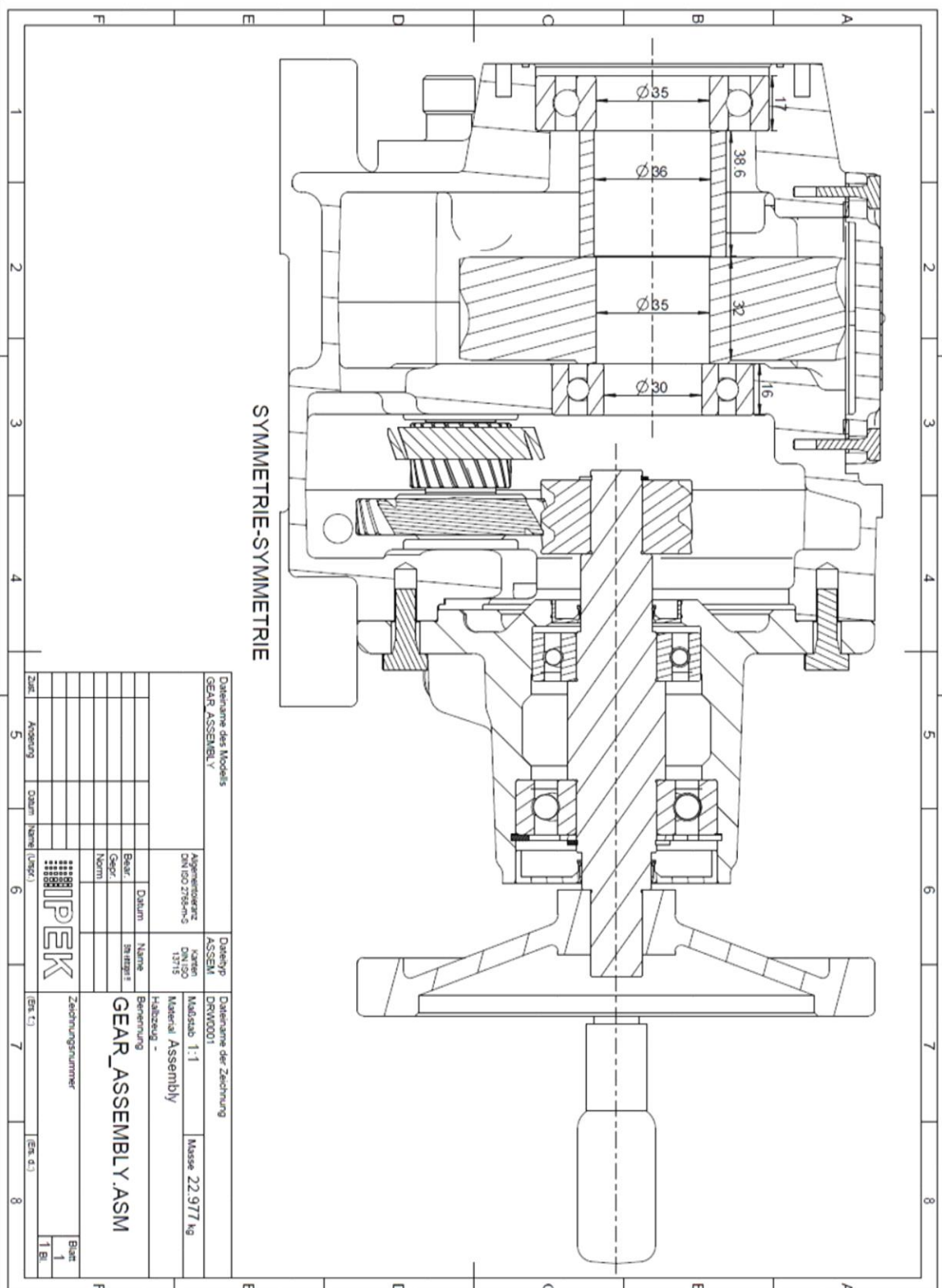


Figure 5: Installation drawing